→ A periodic signal with infinite energy

→ To convert $x(t)$ to finite energy signal: $x(t)e^{-\sigma t}$

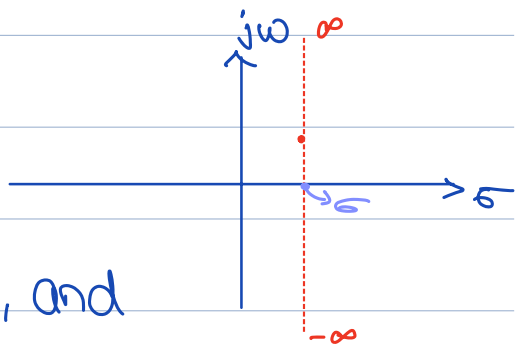
$$X(\sigma+j\omega) = \int_{-\infty}^{\infty} [x(t)e^{-\sigma t}]e^{-j\omega t}\,dt$$

$$\Rightarrow X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}\,dt$$

$$x(t)e^{-\sigma t} = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\sigma+j\omega)e^{j\omega t}\,d\omega$$

$$\Rightarrow x(t) = \frac{1}{2\pi j}\int_{\sigma-j\cdot\infty}^{\sigma+j\cdot\infty} X(s)e^{st}\,ds \qquad s=\sigma+j\omega$$

let $s = 1+j$



fix the real part $= \sigma$, and integrate along a line from $-\infty$ to ∞

$x(t) = e^{10t} \Rightarrow x(t)e^{-\sigma t} = e^{(10-\sigma)t}$ for F.T to exist $\sigma > 10$

Region of convergence

while denoting Laplace transform: $\{X(s), ROC\}$

→ This is not an orthogonal transformation fourier transform was orthogonal whereas

→ with X(s) we can't find energy of x(t) and vice versa since signals aren't orthogonal

$$ROC, x(t) = \{ s : X(s) \text{ exists} \}$$

- $x(t) = e^{-at} u(t) \Rightarrow X(s) = \frac{1}{s+a}$; $Re\{s+a\} > 0$
- For concatenation of signals, ROC is intersection of all the individual ROCs
- If x(t) is right sided and $\sigma = a$ is in ROC, then $\sigma > a$ is also in ROC
- If x(t) is left sided and $\sigma(a)$ is in ROC, then $\sigma < a$ is also in ROC
- If x(t) is finite time duration and bounded, the ROC is entire s-plane
- Pole-zero plot: $X(s) = N(s)/D(s)$ N, D → polynomials

poles: roots of D(s) order of pole = power of root of D(s)

No of poles: degree of D(s)

zeros: roots of NCS order of zero = power of root of NCS

No of zeros: degree of NCS